

# Advanced Topics and Emerging Trends in Generative AI and Assisted Coding

*CSIT205 Generative AI – 2025*



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TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION  
SCIENCES*

*Dr Andrew Yoong, SIM*



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## Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi  
Artist Lauren Henry.



The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, partially cut off. The 'U' is at the top, the 'O' is in the middle, and the 'W' is at the bottom.

# Everyone is welcome here.

UOW takes pride in  
the diversity of our  
community.

Find out more, seek support or join  
the UOW Ally Network  
[uow.info/ally](http://uow.info/ally)

# Subject introduction



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# Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

# Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

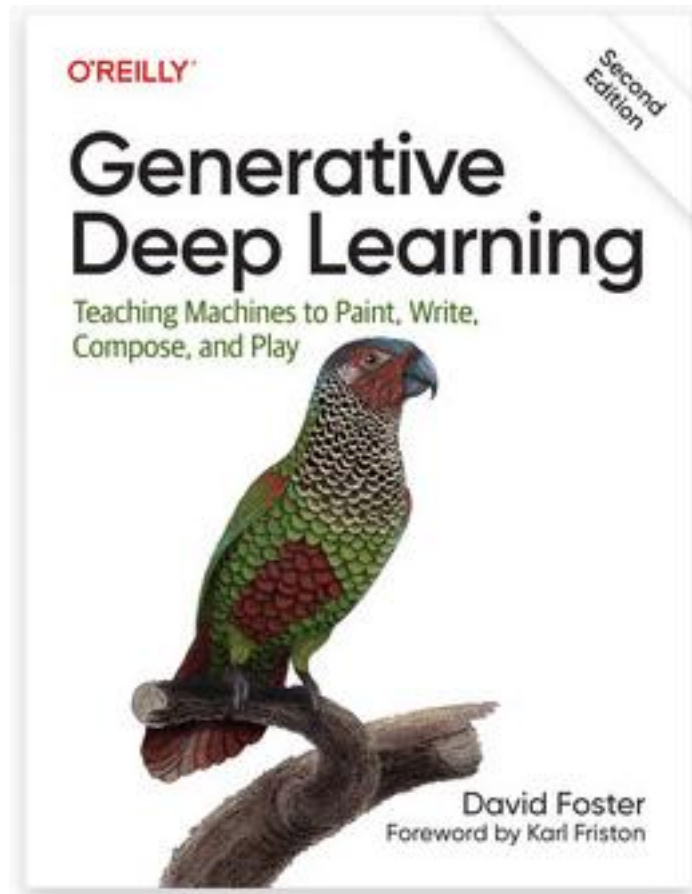
## Week 4: Quiz time

- Week 4's quiz will be open on Moodle after this lecture.
- The quiz is formative (= no official marks = 0% weight).
- Gives you insight into your understanding of Module 1: Generative AI and Assisted Coding (i.e. SLO1).
- It is representative of questions asked during the final exam.



Don't forget 

*This is the book we're using*



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

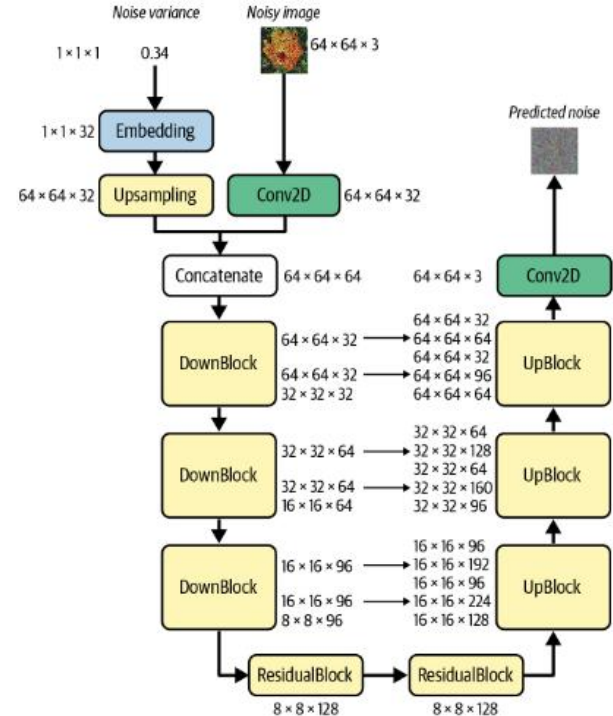
*What did we discuss last week?*



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# Diffusion models

- Unsupervised learning framework
- Uses diffusion and reverse diffusion to learn how to reconstruct an image
- Can learn complex distributions, including those with noise and uncertainty



U-Net (Foster, 2003, p.217)

# What types of AI-assisted coding do we have?



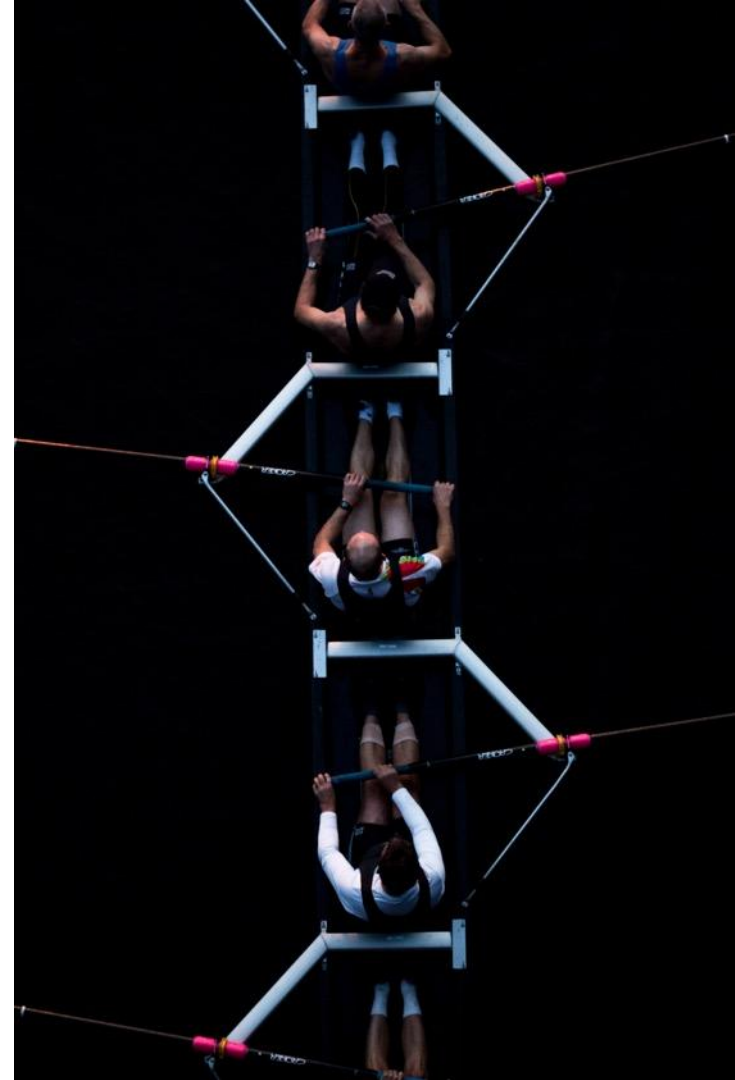
- *Code completion*
- *Code analysis*
- *Code refactoring*

# Why are we here?

*CSIT205 Generative AI - Module 1*

## *Advanced Topics and Emerging Trends*

- *Generative AI*
- *AI-Assisted Coding*





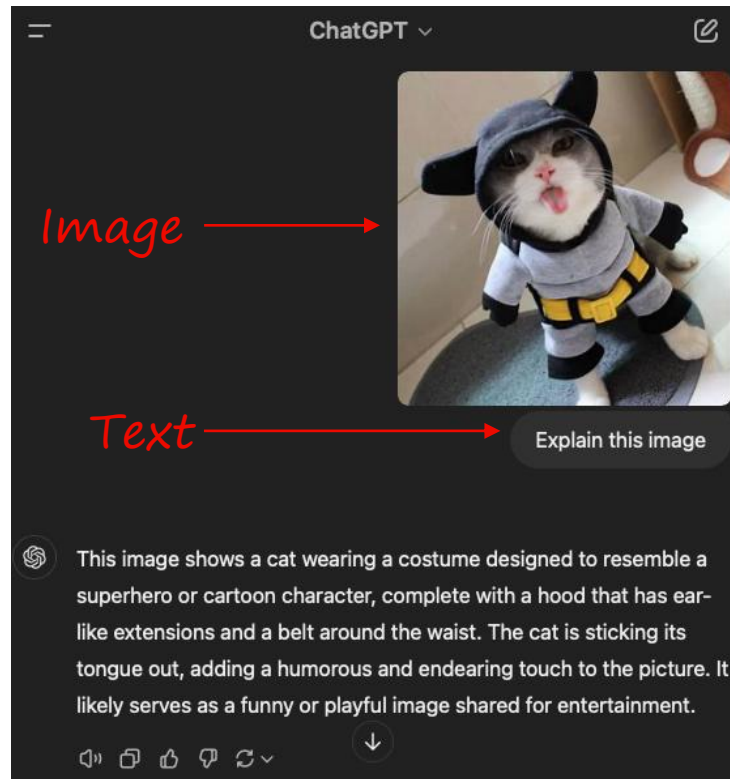
# Trends in Generative AI



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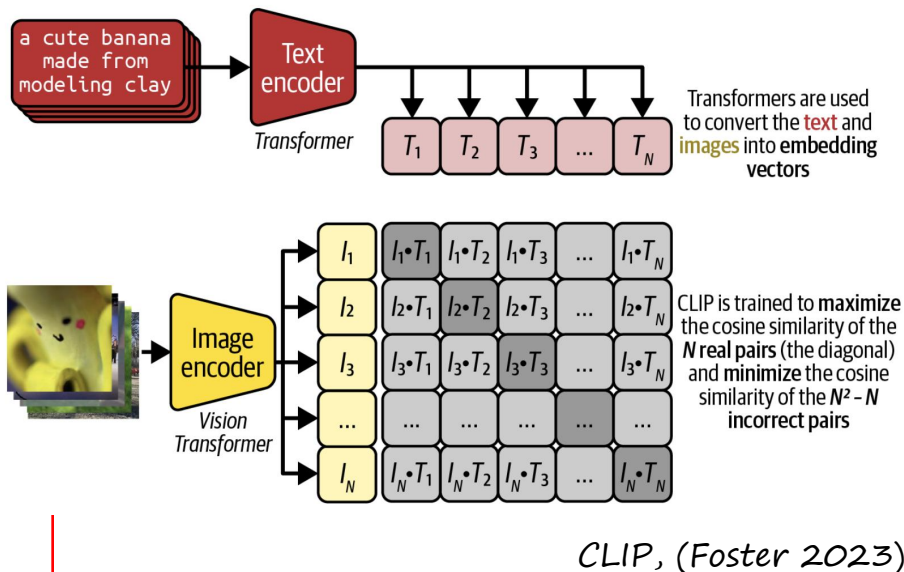
# Multi-modal models

- Multiple modes of input (voice, text, image).
- Examples include Llama 3.2, Gemini, GPT 4o.



# Multi-modal models: text-to-image

- *Text-to-Image and Image-to-Text*
- *Examples include Dall-E 3, Imagen, GPT 4o.*



## Multi-modal models: text-to-video

- *Examples include Dall-E 3, Imagen, GPT 4o.*

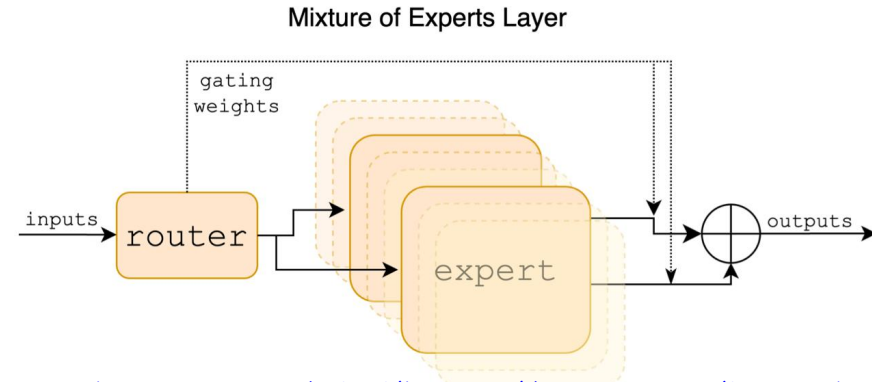
# Model architectures



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# Mixture of Experts (MoE)

- Mixture of Experts (MoE): smaller expert models that each focus on specific sub problems + router that controls experts and output → pre-training with less compute → bigger models + more data!
  - Example: Mixtral 8x7B



<https://www.analyticsvidhya.com/blog/2024/01/discover-the-groundbreaking-llm-development-of-mixtral-8x7b/>

# Challenges with current sequence models

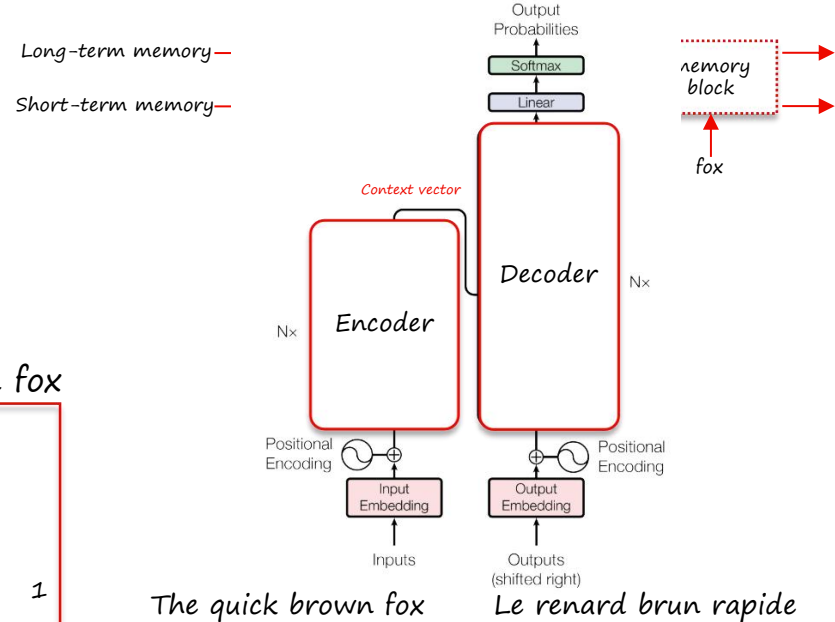
Consider text as a sequence:

- RNNs take into account the whole input sequence (think LSTMs) → longer training
- Transformers apply attention (2D matrix) and convolutions → longer inference

The quick brown fox

|       |     |     |     |   |
|-------|-----|-----|-----|---|
| The   | 1   |     |     |   |
| quick | 0.2 | 1   |     |   |
| brown | 0.3 | 0.2 | 1   |   |
| fox   | 0.4 | 0.6 | 0.7 | 1 |

The quick brown fox





# State Space model intuition

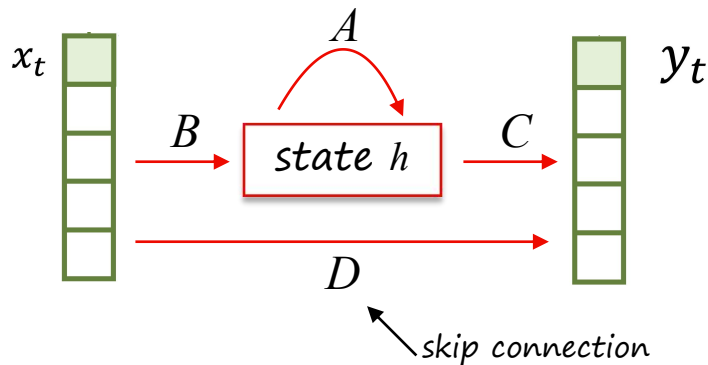
Now consider text as moving from one state (a word) to another state (next word).

💡 No need to model this as a matrix, a vector  $x$  will do.

💡 No need to take the whole input into account, just the previous state  $h$ .

The quick brown fox

state 0 → state 1 → state 2 → state 3



*$A$ ,  $B$ ,  $C$ , and  $D$  are learned parameters*

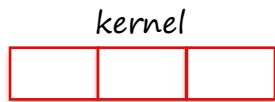
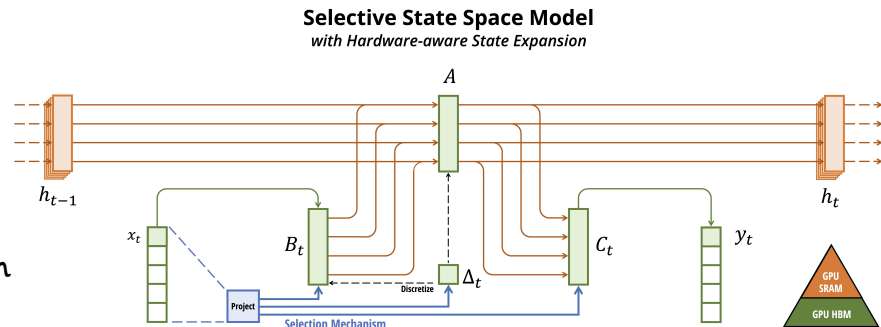
# Mamba / S6 as a Selective State Space Model

💡 Filters out unnecessary input through selective scan operation.

💡 Hardware-ware, because it uses kernel fusion to store state  $h$  in GPU's fast SRAM and  $A$ ,  $B$ , and  $C$  in DRAM.

Simple architecture; Small model (<3B)

- 40x faster training due to parallelism
- 5x faster inference than transformers of similar size.



The quick brown fox

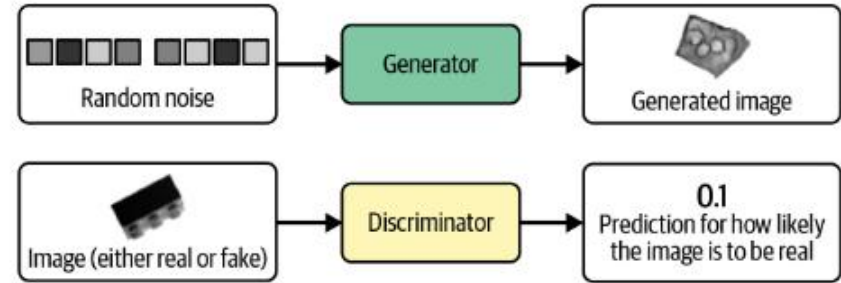
(Don't write intermediate results)

jumps over the lazy dog

<https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mamba-and-state>

# GAN architecture

- *Generator learns to produce realistic data samples by minimising the loss function*
- *Discriminator learns to correctly classify real and fake samples by maximising the loss function*



(Foster, 2003, p.97)

# GAN variants: StyleGAN2 (Karras, 2020)

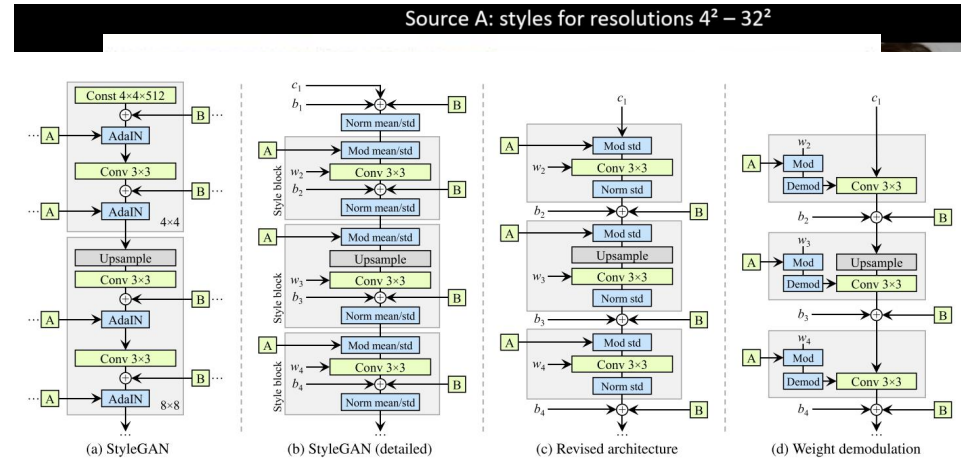
StyleGAN applies a style  $A$  to an image.

However, 'artefacts' (water droplets) may arise due to:

- Architecture: the generator is 'intentionally sneaking signal strength information past normalisation' (Karras, 2020) by creating a local spike.
- Normalisation: applied after noise  $B$  and bias  $b$ , thereby damping their effect.

StyleGAN2 improves this by:

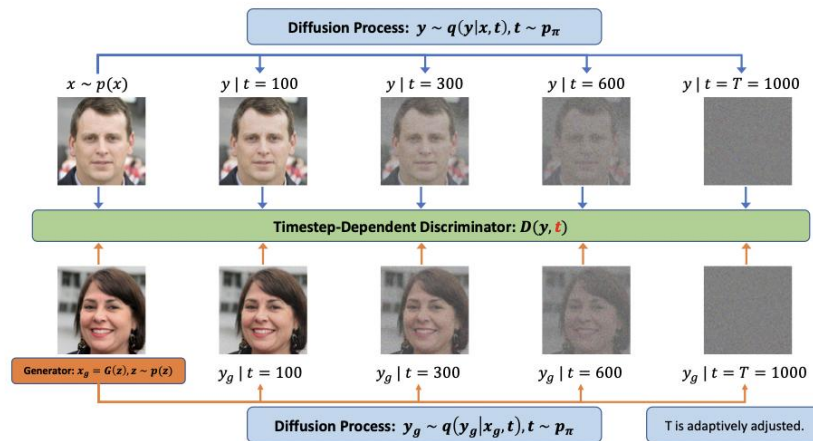
- Moving the bias and noise out of the style block.
- Applying demodulation instead of normalisation.



# GAN variants: Diffusion-GAN

Diffusion-GAN (Wang et al. 2023):  
discriminator compares noisy versions  
instead of the input and generated  
image.

→ easing vanishing gradient problem

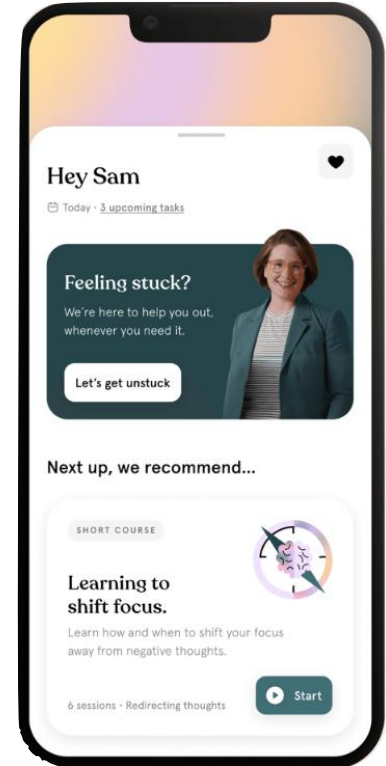


# Domain/task-specific models

*Business applications of generative AI rarely focus on answering questions about the anything on the Web or generating any image.*

*→ More domain-focused models will arise, like:*

- mental health recommendations (Orygen's Mello app)*
- ingredient generation (Canva's Magic AI)*

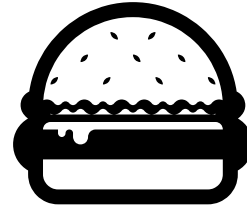


# 10-minute Break!

*This will help you learn*



*10 minutes*



# Fill out the following blanks for Mixture of Experts models



Mixture of Experts (MoE): smaller \_\_\_\_\_ models that each focus on specific sub problem

A. kernel fusion, selective scan

B. router, expert

C. expert, router

D. selective scan, kernel fusion



# What are the two techniques that make Mamba so fast?



- *Selective scan*
- *Kernel fusion*

# What problem(s) does StyleGAN2 solve for StyleGAN that causes water droplets?



*Select all that apply.*

- A. Generator architecture
- B. Normalisation
- C. Modulation
- D. All of the options.

U

Efficiencies

O



W

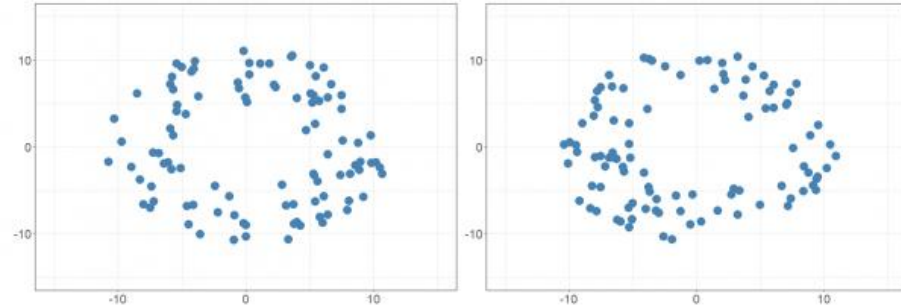
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# Synthetic data

Increasingly strict data privacy rules + data collection effort → new ways to collect data.

Why not generate data using Generative models?

Which model architecture would you use?



Original data

Synthetic data

The synthetic data retains the structure of the original data but is not the same  
<https://datacamp.com/blog/2020/08/20/synthetic-data-unlocking-the-value-of-data-and-privacy-buffer>

# Energy-efficiency

*Large Language Models (LLMs) are expensive in terms of training, fine-tuning and inference compute*

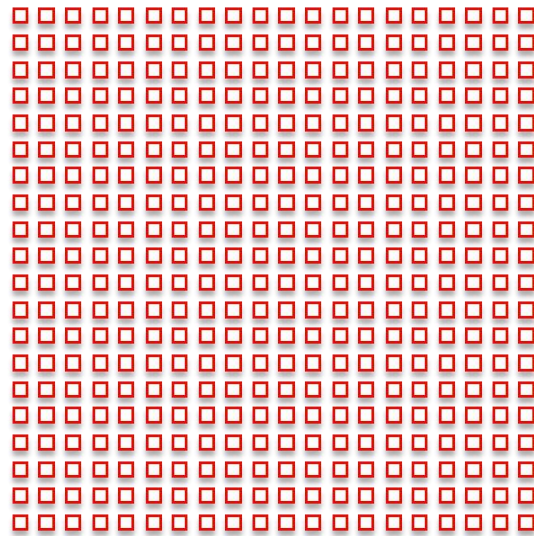
*→ Can we approximate the same performance with Small Language Models (SLMs)? Yes we can!*

*By compressing LLMs, we get SLMs that can run more efficiently, enabling usage on edge devices.*

# Small Language Models

- *Lower Parameter Count: fewer parameters compared to LLMs.*
  - *Faster Inference: sub second vs approx. 5+ seconds.*
  - *Cost-Efficient; can run on edge devices*
  - *Task-Specific: may require fine-tuning to increase task-specific performance.*

Llama 3.1 405B (LLM)

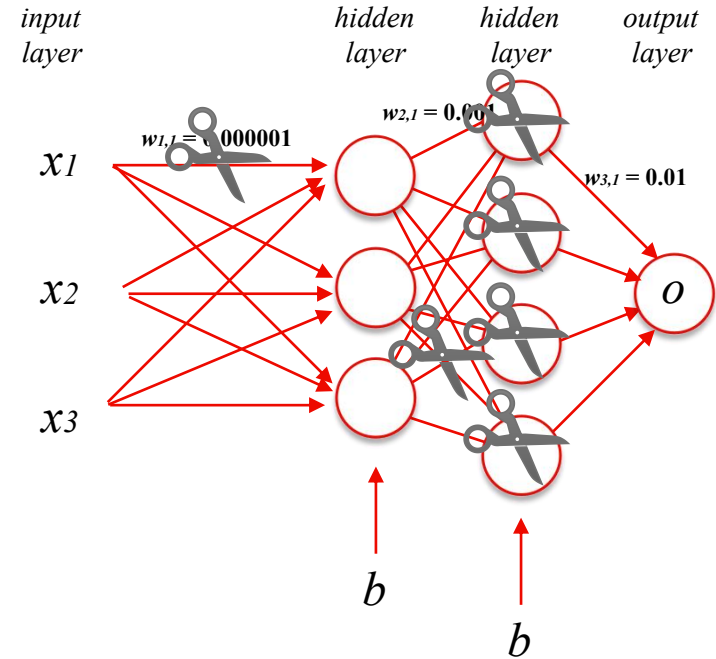


Llama 3.2 1B (SLM)



# How are Small Language Models created?

- Pruning: cutting (near-)zero weights from the neural network.
- Quantization: converting weights from 32-floating point numbers to for instance 8-bit integers or [bfloat16](#).
- Knowledge distillation: train a small 'student' model to learn behaviour similar to a bigger 'teacher' model (e.g. ChatGPT).
- Parameter efficient fine-tuning (to be discussed later)

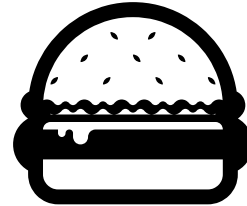


# 10-minute Break!

*This will help you learn*



*10 minutes*





# What are three techniques to create small language models?



- *Pruning*
- *Quantization*
- *Knowledge distillation*

# Trends in AI-assisted Coding



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# Autonomous AI assisted coding

- *Write code and generate a pull request*
- *Proactively identify and address issues: create / run tests and solve issues.*
- *Code migration, e.g. from Java to Python, run version upgrades*
- *Run CI/CD pipelines*

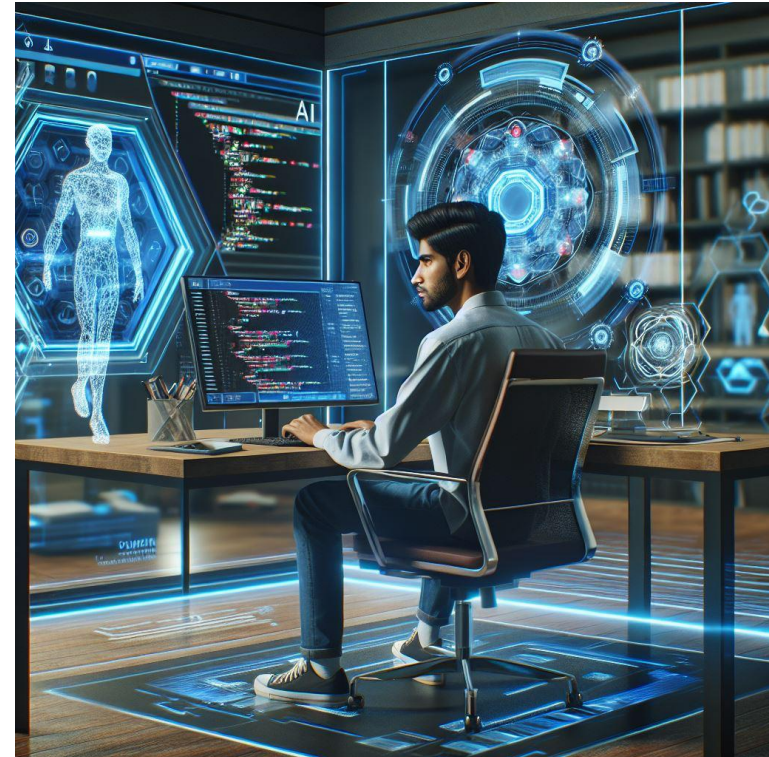
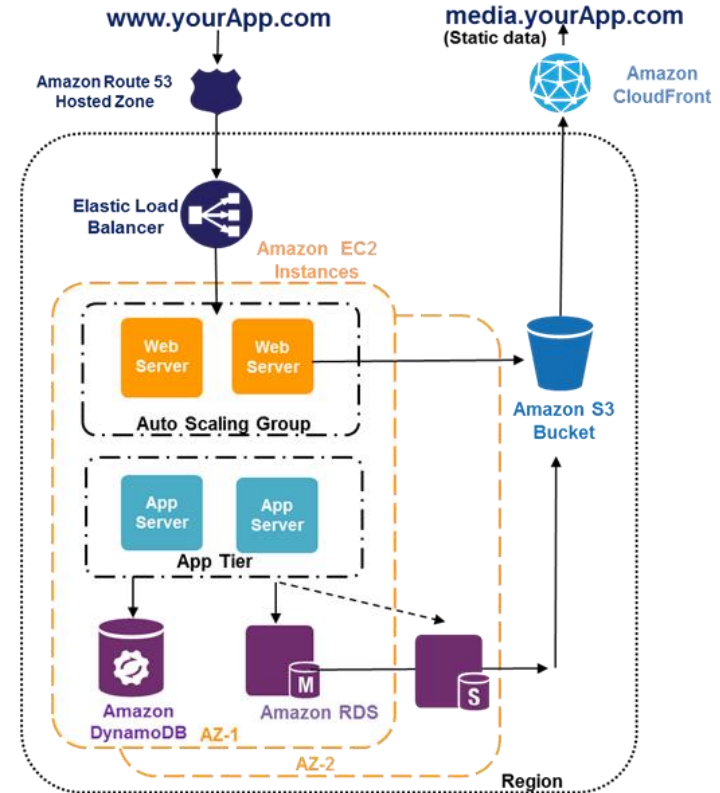


image of a futuristic coding environment with AI-inspired tools and interfaces, MS Copilot

# Multi-modal assisted coding

- *Provide a diagram (image) and prompt (text) the model to write the code for you.*
- *Critique the diagram.*



# Security

Automated repairs of Common Weakness Enumeration (CWE): LLM4CVE (Fakih, et al. 2025) uses:

- LLM (e.g. Llama 3 70B)
- Prompt:
  - Potentially vulnerable code
  - CWE
- Automated feedback on code similarity and not introducing new vulnerabilities

```
1 unsigned char * snmp_ber_decode_type(  
    unsigned char *buff, uint32_t *buff_len,  
    uint8_t *type)  
2 {  
3     if(*buff_len == 0) {  
4         return NULL;  
5     }  
6  
7     *type = *buff++;  
8     (*buff_len)--;  
9  
0     return buff;  
1 }
```

Listing 1: The original vulnerable function

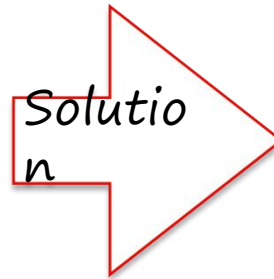
```
1 unsigned char* snmp_ber_decode_type(unsigned  
    char *buff, uint32_t *buff_len, uint8_t  
    *type) {  
2     if(buff == NULL || buff_len == NULL ||  
        type == NULL || *buff_len == 0) {  
3         return NULL;  
4     }  
5  
6     *type = *buff++;  
7     (*buff_len)--;  
8  
9     return buff;
```

Code repair snippet from (Fakih, et al. 2025)

# Where to from here?

*Gergely Orosz (The Pragmatic Engineer) and Addy Osmani (Google) on how AI will be used:*

- *Bootstrappers vs iterators*
- *Use it to automate / accelerate routine tasks*
  - *But: Verify the results*
- *Quickly get a grasp of concepts*
  - *But: you'll miss out on fundamental skills, like debugging, architecture, design patterns.*



*Patterns:*

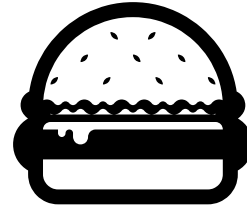
- *AI first draft*
- *Constant conversation*
- *Trust but verify*

# 10-minute Break!

*This will help you learn*



*10 minutes*



# What are the three patterns to safely use AI-assisted coding?



- *AI first draft*
- *Constant conversation*
- *Trust but verify*



# Wrap-up



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# Summary

*Trends in Generative AI:*

- *Mixture of Experts*
- *SG Mamba*
- *Multi-modal models*

*Trends in Assisted Coding:*

- *autonomous agents,*
- *multi-modal assisted coding*
- *security*

# For next lecture

- Read Wang et al., 2023; Wei et al., 2022 as preparation for Module 2: Prompt Engineering.
- Try the quiz!
  - Formative Assessment: 0% weight towards final mark/grade
  - To test your knowledge of Module 1
  - 📁 Will help you prepare for the final exam

# References and Resources

- Fakih, M., Dharmaji, R., Bouzidi, H., Araya, G. Q., Ogundare, O., & Faruque, M. A. al. (2025). LLM4CVE: Enabling Iterative Automated Vulnerability Repair with Large Language Models. <http://arxiv.org/abs/2501.03446>
- Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J., & Aila, T. (2020). Analyzing and improving the image quality of stylegan. Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 8107–8116. <https://doi.org/10.1109/CVPR42600.2020.00813>
- Schick, T., & Schütze, H. (2021). It's Not Just Size That Matters: Small Language Models Are Also Few-Shot Learners. Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2339–2352. <https://doi.org/10.18653/v1/2021.naacl-main.185>
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- Wang, S., Kanwar, P. (2019). BFloat16: The secret to high performance on Cloud TPUs. <https://cloud.google.com/blog/products/ai-machine-learning/bfloat16-the-secret-to-high-performance-on-cloud-tpus>
- Wang, Z., Zheng, H., He, P., Chen, W., & Zhou, M. (2022). Diffusion-GAN: Training GANs with Diffusion. <http://arxiv.org/abs/2206.02262>
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. v., & Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. Advances in Neural Information Processing Systems, 35. <https://arxiv.org/abs/2201.11903v6>

# Questions



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